PROJECT · EMBEDDED SYSTEMS

Smart parking with biometric access

Polytechnique Sousse · 2024/25 · team of 2

2-factor
fingerprint, then PIN keypad

Arduino Uno
C++ control logic

5 places
live count on an I²C LCD

CONTEXT

A small parking prototype that only lets registered users in and always shows how many places are free.

WHAT I DID

- Entry: fingerprint match, then a 4-digit code, then a servo barrier opens.
- Exit: a PIR motion sensor detects the car and opens the second barrier.
- The place counter updates on every entry and exit and is shown on the LCD.
- Integrated the fingerprint module, keypad, PIR, two servos and the LCD around an Arduino Uno.

TOOLS AND METHODS

Arduino · C++ · Fingerprint sensor · PIR · Servo · LCD I²C

OUTCOME

Planned next steps: several user codes, notifications, and a camera with plate recognition.

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